## Conjugation by diag(1, −1) on $M_2(F_3)$

| coordinate | $T - I$ | N for $C_2$ | N for $C_6$ |
|---|---|---|---|
| $E_{11}$ | 0 | 2 | 0 |
| $E_{12}$ | 1 | 0 | 0 |
| $E_{21}$ | 1 | 0 | 0 |
| $E_{22}$ | 0 | 2 | 0 |

$C_2$: dim ker N = 2
dim $H^1$ = 0

$C_6$: dim ker N = 4
dim $H^1$ = 2